



Simulations of firebrand-driven fire spread in landscape-scale Wildland-Urban-Interface (WUI) and urban conflagration models

Yiren Qin^{a,b}, Dwi M.J. Purnomo^b, Maria Theodori^b, Maryam Zamanialaei^b,
Chris Lautenberger^c, Michael Gollner^b, Arnaud Trouvé^{a,*} 

^a Department of Fire Protection Engineering, University of Maryland, College Park, MD, 20742, USA

^b Department of Mechanical Engineering, University of California, Berkeley, CA, 94720, USA

^c CloudFire Inc., 985 Lincoln Way, Suite 207, Auburn, CA, 95603, USA

ARTICLE INFO

Keywords:

Structure-to-structure fire spread
WUI fires
Urban conflagrations
Landscape-scale fire modeling
Firebrand
Ember
Spot fire ignition
Fire risk modeling

ABSTRACT

This study presents an application of a recently developed firebrand model to the case of fire spread in WUI and urban communities using a landscape-scale formulation. The study considers a benchmark configuration corresponding to an array of identical, equally-spaced, square structures; the array is characterized by two length scales: the structure size (SS) and structure separation distance (SSD). Simulations are performed both in one-dimension using MATLAB and in two-dimension using an established landscape-scale fire spread simulator called ELMFIRE. The firebrand model features descriptions of generation, transport and ignition: these were recently introduced for wildland fire applications and are here adapted to the case of WUI/urban fires. Simulations provide an evaluation of the critical conditions that lead to structure-to-structure fire spread, in terms of SSD, wind velocity and firebrand generation rates. The study also includes an evaluation of the model sensitivity to changes in spatial resolution. The model is found to be sensitive to changes in spatial resolution and requires a resolution of less than SS and SSD for grid convergence. These results can be explained by the dominant role of SS and SSD in the WUI/urban fire problem: both length scales need to be captured by the computational grid.

1. Introduction

Fire behavior in general, whether indoor or outdoor, features multi-physics phenomena in the form of complex interactions between thermal degradation of combustible solids, turbulent flow, combustion, and convective/radiative heat transfer. At the Wildland Urban Interface (WUI), or inside urban areas, these multi-physics phenomena occur at multiple scales. It is useful to differentiate between the compartment/individual-building scales, $L_{building}$, that correspond to indoor fire behavior and object-to-object and room-to-room fire spread; the building cluster scales, $L_{cluster}$, that correspond to outdoor fire behavior and structure-to-structure fire spread; and the landscape/community scales, $L_{landscape}$, that correspond to fire propagation across an entire WUI/urban community, including the interaction of the fire with local topography and wind conditions. In addition, the flame has scales that can be represented by a characteristic length, L_{flame} ; the flame scales take a large range of values as they grow from small indoor flames confined

to a room to large outdoor flames that may spread across several structures. Furthermore, the fire plume has scales that can be represented by a characteristic length and width, L_{plume} and W_{plume} ; the plume scales take a large range of values as they grow from building scales to cluster scales and then to landscape scales. In WUI/urban fire problems, $L_{building}$ is meter-scale; $L_{cluster}$ is 10-m-scale; and $L_{landscape}$ is kilometer-scale and beyond.

Because WUI/urban fires are complex multi-physics multi-scale phenomena, computational tools are required to make engineering predictions of their global properties, e.g., their rate of spread, rate of heat release, flame height, resulting structure damage, emission yields, etc. Because of the large spectrum of relevant scales, the computational models used to simulate wildland fires are typically limited to a specific range of scales; one differentiates between: (1) “building fire solvers” that resolve indoor fire dynamics at room and building scales; (2) “building cluster fire solvers” that resolve outdoor fire dynamics at multiple building scales; and (3) “urban conflagration solvers” that

This article is part of a special issue entitled: FISJ IAFSS 2026 published in Fire Safety Journal.

* Corresponding author.

E-mail addresses: yirenqin712@berkeley.edu (Y. Qin), dwimjpurnomo@berkeley.edu (D.M.J. Purnomo), theodori@berkeley.edu (M. Theodori), zamanialaeim@berkeley.edu (M. Zamanialaei), chris@cloudfire.com (C. Lautenberger), mgollner@berkeley.edu (M. Gollner), atrouve@umd.edu (A. Trouvé).

<https://doi.org/10.1016/j.firesaf.2026.104686>

Received 23 September 2025; Received in revised form 11 February 2026; Accepted 3 March 2026

Available online 6 March 2026

0379-7112/© 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC license (<http://creativecommons.org/licenses/by-nc/4.0/>).

resolve outdoor fire dynamics at landscape/community scale. Currently, “building fire solvers” and “building cluster fire solvers” are based on classical first-principles Computational Fluid Dynamics (CFD) approaches, while “urban conflagration solvers” are “non-CFD” solvers.

Examples of “building fire solvers” include FDS [1], OpenFOAM [2] and FireFOAM [3]. FireFOAM is a specialized version of OpenFOAM developed by FM for building fire applications. These solvers use a computational grid resolution of order 1–10 cm and provide a fine-grained treatment of the pyrolysis, combustion and heat transfer processes that are responsible for fire spread through a first-principles-based approach.

Examples of “building cluster fire solvers” include FDS [1] and WFDS [4]. These solvers use a computational grid resolution of order 1 m and provide a coarse-grained treatment of unresolved building-scale processes through a simplified (but physics-based) design fire curve model. “Building cluster fire solvers” include a description of the three main mechanisms responsible for structure-to-structure heat transfer and fire spread: (1) radiation heat transfer; (2) convective heat transfer (also called “flame or plume contact”); and (3) firebrands.

Examples of “urban conflagration solvers” currently being developed in the U.S. for WUI/urban fire applications include SWUIFT [5,6], AGNI-NAR [7–9], and ELMFIRE [10,11]. Other examples of “urban conflagration solvers” include WUI fire models [12–15], post-earthquake fire models [16,17] and models developed to evaluate the risk of fire in informal-settlements [18,19]. Ref. [20] provides an excellent review of “urban conflagration solvers”. These solvers use low spatial resolution (typically 30 m), feature different flavors of semi-empirical, deterministic or probabilistic models to describe fire spread, and are informed by detailed descriptions of the vegetation maps, the building maps, topographic maps and weather conditions. The solvers are “non-CFD”: they rely on a variety of computational techniques, including for instance graph theory [7–9], cellular automata [13–15], level-set methods [10,11], and/or simplified descriptions of the repeated heating/ignition/combustion cycle that characterizes fire spread [5,6,10,11,16–19]. The solvers are used primarily as models to calculate the fire risk in WUI/urban communities; they can also be used as operational models during response to fire events. The models currently have large uncertainties, in part because validation data from past fire disasters are typically limited to fire perimeters obtained at low spatial and temporal resolution, and to post-fire observations on structure damage.

The present study considers the landscape-scale modeling approach adopted in “urban conflagration solvers”. The study is a continuation of previous work aimed at extending the modeling capability of an established landscape-scale fire spread simulator called ELMFIRE [21,22] with the objective to simulate firebrand-driven fire spread [23,24]. Firebrands are reported to be a key factor, and often the dominant factor, controlling structure-to-structure fire spread during WUI fires and urban conflagrations [25–28]. Basic processes of interest in firebrand-driven WUI/urban fire spread include: the generation of firebrands by a burning “source structure”; their transport (in flaming or smoldering state) by the fire plume and the local wind to distant landing locations, possibly close to one “target structure”; and the subsequent ignition of a spot fire at the exposed target structure, possibly involving surface creep motions of the firebrands, ground accumulation around obstacles, smoldering-to-flaming transition, and transition from an outdoor fire to an indoor fire. The new firebrand model presented in Refs. [23,24] includes descriptions of generation, transport and ignition: these model descriptions were first tested in wildland fire applications and are here further evaluated and adapted to the case of WUI/urban fires.

Similar to previous work in Refs. [23,24], the present study adopts a model development strategy based on simple test configurations with simulations performed first using MATLAB for ease of analysis and then reproduced using ELMFIRE. Section 2 presents the main features of the firebrand model. Section 3 presents results obtained in a benchmark

configuration corresponding to a one-dimensional array of identical, equally-spaced, square structures; simulations are performed with MATLAB and structure-to-structure fire spread is simulated for different values of the structure separation distance, wind velocity and firebrand generation rate. Note that the present test configuration only considers fire spread due to firebrands; spread mechanisms due to thermal radiation and flame/plume contact are here ignored (the ELMFIRE treatment of these other fire spread mechanisms is the focus of separate studies, see Refs. [10,11]). Section 4 presents results obtained in the same benchmark configuration except that the configuration is now two-dimensional and that simulations are performed with ELMFIRE. Both sections 3 and 4 include a novel evaluation of the sensitivity of the firebrand model to changes in spatial resolution. This evaluation reveals a surprisingly strong sensitivity, which suggests that the role of spatial resolution has been largely overlooked in past studies and that the treatment of fire spread in WUI/urban communities is more demanding than that of fire spread in the wildland.

It is important to emphasize that the focus of the present study is on verification tests of the firebrand model (rather than validation tests). The objective of verification tests is to establish the ability of a model to provide predictions that are free of numerical errors (in contrast, the objective of validation tests is to establish the ability of a model to provide predictions that are accurate representations of physical reality.) Consistent with this objective, the numerical configuration corresponds to an idealized academic configuration featuring an array of identical, equally-spaced, square structures, constant and uniform wind, and flat terrain. The firebrand model is not evaluated against data corresponding to real conflagrations.

2. Firebrand model

This section presents a brief description of the firebrand model proposed in Refs. [23,24].

ELMFIRE uses a two-dimensional level-set model to describe fire spread due to thermal radiation and flame/plume contact. In the case of a horizontal flat terrain, the (x - y) computational domain is the horizontal plane; in the case of more complex terrain, the (x - y) computational domain is a reference horizontal map that corresponds to projected variations. This baseline treatment requires the calculation of the rate of spread of the fire, ROS , which is obtained through a physics-based model that contains information on wildland and WUI/urban fire behavior [10,11,21,22].

The level-set model in ELMFIRE has been adapted in Refs. [23,24] to include discrete events associated with the ignition of spot fires. The spot fire model features: a description of the landing locations of firebrands, (x_{ck}, y_{ck}) , where k is the index of a computational cell in the discretization of the (x - y) domain; the time of arrival of the firebrands, TOA_k ; and the ignition delay, $t_{\text{ignition},k}$.

The firebrand transport algorithm calculates (x_{ck}, y_{ck}) based on a prescribed probability density function (PDF) of flight distance [23]. The PDF of firebrand flight distance is described in terms of the wind speed, u_{wind} , and the heat release rate of the fire, HRR , using expressions proposed by Sardoy et al. for burning vegetation [29] and by Himoto & Tanaka for burning structures [30]. In the case of a vegetation fire, HRR is calculated from the model for the rate of spread, ROS ; in the case of a WUI/urban fire, HRR is obtained from a design fire curve that describes the temporal variations of the burning intensity as they apply to structure fires from ignition to burnout and without a suppression response. The literature suggests peak values of HRR of several tens of MW and a duration from several tens of minutes up to several hours [13,31].

Fig. 1 presents the design fire curve selected in this study for an assumed $(10 \times 10 \text{ m}^2)$ structure. The fire curve features three phases: a linear growth phase that lasts 300 s; a 1-h-long fully-developed phase with a constant peak value of HRR equal to 40 MW; and a linear decay phase that lasts 300 s. This fire curve is consistent with choices made in

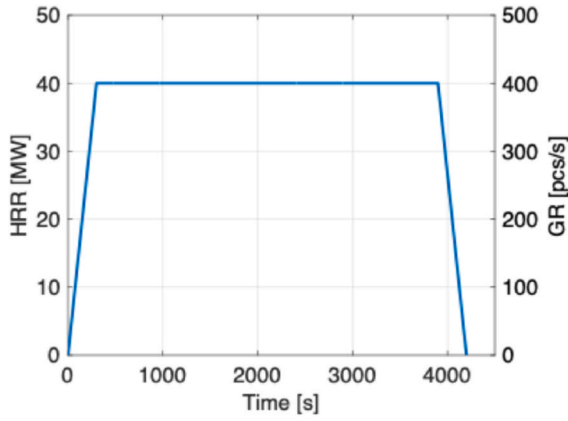


Fig. 1. Prescribed temporal variations of the heat release rate, HRR (left y-axis), and the firebrand generation rate, GR (right y-axis), in a burning structure. In this plot, time $t = 0$ corresponds to the transition to a fully-developed structure fire.

earlier work on WUI fires performed with ELMFIRE [10,11]. The peak value of the heat release rate is based on surveys of fire loads and estimates of burning behavior presented in Refs. [13,31]; our choices correspond to a fire load of 1.56 GJ/m^2 and a peak burning rate of 400 kW/m^2 (both quantities are given per unit square-meter of the structure footprint). Note that the selected design fire curve presented in Fig. 1, while realistic, does not correspond to a particular representation; the choice of a design fire curve is an important step and is made on a case-by-case basis in applications of ELMFIRE to WUI/urban fires. The plot in Fig. 1 also contains information on the firebrand generation rate, GR , to be discussed below.

Fig. 2 presents representative PDFs of firebrand flight distance in the downwind direction. These PDFs correspond to high wind conditions, $U_{wind} = 17.9 \text{ m/s}$ (40 mph), and to firebrands emitted from burning structures with $HRR = 10, 40$ or 160 MW . The PDFs use the lognormal distributions proposed in Ref. [30]; in the firebrand model, the PDFs are discretized and truncated at the 99th percentile to a maximum value of flight distance, noted X_{max} [23]; we have $X_{max} = 65 \text{ m}$ (10 MW), 84 m (40 MW), and 109 m (160 MW). These values indicate that the focus of the model is on short-distance firebrands. Note that in Ref. [30], the PDFs of firebrand flight distance do not depend explicitly on firebrand shape and weight. In the following, consistent with the choice made in Ref. [23], the average firebrand weight is assumed to be 0.2 g . Also, the firebrands considered in the model correspond to firebrands

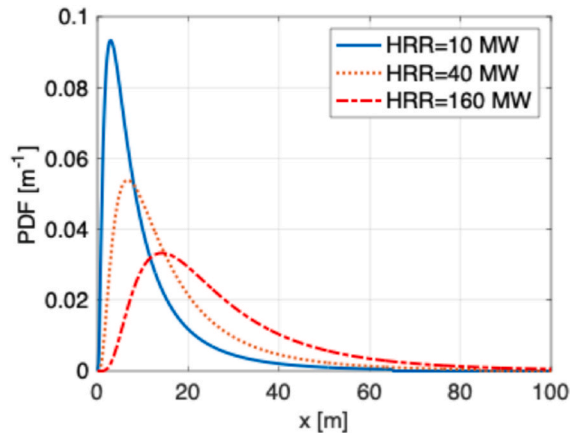


Fig. 2. Probability density function (PDF) of firebrand flight distance for firebrands emitted by a burning structure with $HRR = 10, 40$ or 160 MW and transported downwind by a 40-mph wind, $u_{wind} = 17.9 \text{ m/s}$. The PDFs are taken from Ref. [30].

that are in flaming or smoldering state when landing on the ground.

The firebrand transport algorithm also calculates TOA_k based on a simplified model for the flight time of firebrands [23]. The model currently assumes that firebrands move horizontally at the speed of the wind evaluated at 6.1-m elevation (the model neglects variations in elevation due to lofting or gravity, a simplification that will be removed in future work).

The firebrand ignition model calculates $t_{ignition,k}$ based on a description that considers two scenarios: in wildland fires, the model assumes that spot ignition results from the landing of a single firebrand in flaming state; in contrast, in WUI/urban fires, the model assumes that structure ignition results from the ground accumulation of firebrands in smoldering state and can be characterized by two consecutive time delays, a first delay, $t_{ign,small,k}$, associated with the occurrence of a small flame due to smoldering-to-flaming transition, and a second delay, $t_{ign,large,k}$, that corresponds to the transition from a small flame to a fully-developed fire [24].

Focusing on the WUI/urban fire scenario, the model for $t_{ign,small,k}$ is based on an empirically-derived probability of ignition proposed by De Beer et al. [32,33] and is described as a function of the accumulated mass of firebrands per unit ground surface, ψ_k (in units of g/cm^2), and the wind velocity, $v_{air,k}$ (measured at near-ground elevation and obtained from the wind velocity at 6.1-m elevation through an assumed velocity boundary layer profile; we use a profile proposed in Ref. [34]; note that the model does not account for expected modifications of the local wind by structures and burning fuels.). Fig. 3 presents the probability of small flame ignition adapted from the approach proposed by De Beer et al. [32,33] and discussed in Refs. [24,35]. The present map corresponds to pressure-treated wood with a moisture content equal to 2.0%. One finds that small flame ignition resulting from smoldering-to-flaming transition is essentially a binary process: for given values of v_{air} and ψ , the probability of ignition is either close to 0 (failed ignition) or close to 1 (successful ignition). In Fig. 3, the critical ignition conditions are approximated by an analytical curve that is fitted to the contour $P_{ign,small} = 0.5$:

$$\psi^*(v_{air}) = \frac{C}{(v_{air} - v_{min})(v_{max} - v_{air})} \quad (1)$$

where $v_{min} = -0.073 \text{ m/s}$, $v_{max} = 4.111 \text{ m/s}$ and $C = 0.211$, and where

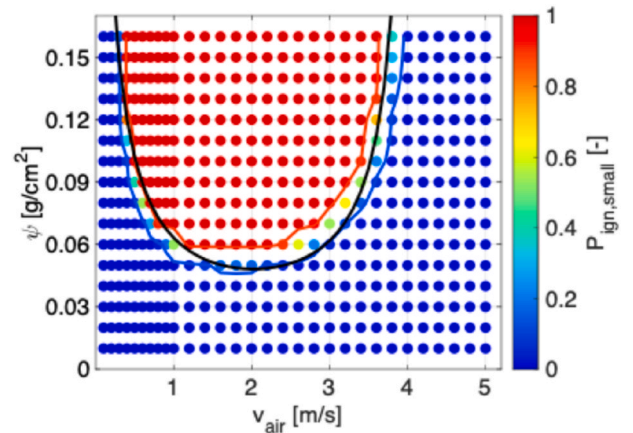


Fig. 3. Probability of small flame ignition as a function of the near-ground wind velocity, v_{air} , and the accumulated mass of firebrands per unit ground surface, ψ . The model follows the approach proposed by De Beer et al. [32,33] and is presented in Ref. [24]. The present map corresponds to pressure-treated wood (PTW) with a moisture content equal to 2.0%. The blue (red) solid line corresponds to $P_{ign,small} = 0.1$ (0.9); the black solid line is a fitted curve that approximates the contour $P_{ign,small} = 0.5$ (see Eq. (1)). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

the subscript k has been dropped for clarity. The function in Eq. (1) has a U-shape and two asymptotic lines at $v_{air} = v_{min}$ and $v_{air} = v_{max}$ (the negative value of v_{min} results from regression analysis and does not necessarily carry physical meaning). The ignition model can now be recast as follows: successful ignition corresponds to the region $\psi \geq \psi^*(v_{air})$, while failed ignition corresponds to the region $\psi < \psi^*(v_{air})$. Eq. (1) suggest that ignition will not occur for critically low or high values of v_{air} , and for critically low values of ψ . Critically low values of v_{air} do not provide sufficient mass transfer of oxygen for the smoldering reaction; critically high values of v_{air} provide excessive convective cooling; critically low values of ψ do not provide sufficient radiative heating.

The ignition delay $t_{ign,small,k}$ represents the time to collect enough firebrands at a target structure plus the time to transition to flaming combustion, 42 s in Ref. [24]. The model for $t_{ign,small,k}$ also requires an algorithm that relates the probability of small flame ignition presented in Fig. 3 to an ignition time scale; we refer the reader to Ref. [24] for a description of this algorithm.

The model for $t_{ign,large,k}$ is based on a t -square law and describes the time required for the growth from a small flame (often an outdoor flame) due to smoldering-to-flaming transition to a large flame (now an indoor flame) that is sufficiently large to sustain fire spread (inside the target structure) [24]. Following Ref. [24], we use $t_{ign,large,k} = 300$ s. Note that the target material (pressure-treated wood - PTW) and moisture content are fixed and are not varied in the present study. The model assumes that PTW (not zone-0 fuel) will be ignited by firebrands. Future work will have to consider variations in target material, moisture content and fire penetration scenarios.

The description of the ground accumulation of firebrand mass, ψ_k , used in the firebrand ignition model requires a firebrand generation model. The firebrand generation model uses an empirical relationship in which the generation rate of hot firebrands, GR , is treated as proportional to the heat release rate of the fire, HRR [24]. The literature suggests that the coefficient of proportionality takes values between 10 and 100 pcs/s/MW for burning vegetation [36and10] pcs/s/MW for burning structures [37]. Fig. 1 presents the values of GR associated with the design fire curve used in the present study: these values assume a generation rate per unit HRR equal to 10 pcs/s/MW.

3. One-dimensional structure-to-structure fire spread in MATLAB

We now turn to an evaluation of the performance of the firebrand model and consider a benchmark configuration corresponding to an array of identical, equally-spaced, square-shaped, one-story structures; the array is characterized by two (horizontal) length scales: the structure size (SS) and the structure separation distance (SSD). The baseline configuration corresponds to $SS = SSD = 10$ m. In Section 3, the configuration is one-dimensional and simulations are performed using MATLAB for ease of analysis. In Section 4, the configuration is two-dimensional and simulations are performed using ELMFIRE.

3.1. Numerical configuration

Fig. 4 presents the numerical configuration for different representative levels of spatial resolution, as characterized by Δx , where Δx is the size of a computational cell (the computational grid is uniform) and where x designates the wind direction. We consider here 15 different levels of spatial resolution ranging from $\Delta x = 1$ m to $\Delta x = 50$ m (Fig. 4 presents 3 of them); simulations with $\Delta x < SS, SSD$ correspond to fine resolution, while simulations with $\Delta x \geq SS, SSD$ correspond to coarse resolution. The middle figure in Fig. 4 corresponds to $\Delta x = 10$ m and shows the exact geometry of the problem: the structures have a dimension $SS = 10$ m; the spaces that separate them have a dimension $SSD = 10$ m; these spaces are non-combustible (and therefore do not emit firebrands) and act as firebreaks. For each value of Δx , the

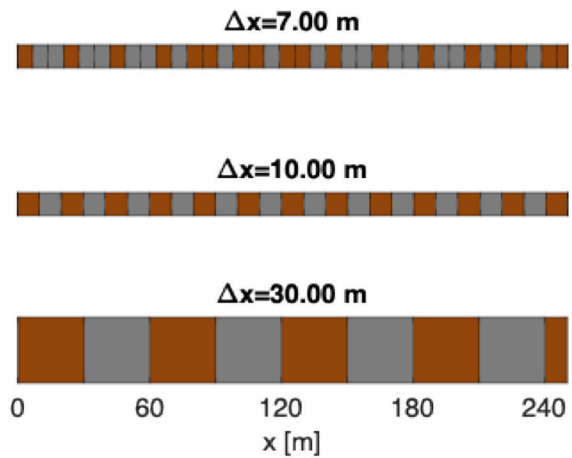


Fig. 4. Numerical configuration used in the one-dimensional MATLAB tests and corresponding to an array of alternating structures (colored in brown) and firebreaks (colored in grey). The top (bottom) figure corresponds to fine-resolution (coarse-resolution), $\Delta x = 7$ m ($\Delta x = 30$ m); the middle figure corresponds to $\Delta x = 10$ m and shows the exact geometry of the problem in which $SS = SSD = 10$ m. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

numerical layout of structures is determined using a mode-resampling method, in which each cell at the target resolution covering portions of structures and portions of firebreaks is defined as a structure or a firebreak based on which attribute has the largest portion. The implementation of the algorithm in MATLAB follows the same procedure as that in GDAL [38], a widely used Geographic Information System (GIS) pre-processing package and the current data pre-processor used by ELMFIRE.

Fig. 4 shows that at high spatial resolution ($\Delta x < 10$ m), the geometrical length scales SS and SSD are approximately captured by the computational grid and their effective values are close to the true values: for instance for $\Delta x = 7$ m, the effective size of structures is equal to 7 or 14 m; the effective separation distance is also equal to 7 or 14 m (higher levels of spatial resolution would decrease these inaccuracies). The 14-m structures obtained with $\Delta x = 7$ m may be treated in two different ways: as a single unit (i.e., one 14-m-size unit) characterized by a value of the heat release rate that corresponds to the whole unit and an emission point of firebrands that is located at the center of gravity of the unit; or as multiple discrete cells (i.e., two 7-m-size cells) characterized by values of the heat release rate and emission points of firebrands that are distributed and defined on a per-cell basis. The first treatment is more accurate since it correctly recognizes structures as the relevant unit in the fire spread problem; the second treatment is less accurate but is the only treatment currently available in ELMFIRE. We consider both treatments in MATLAB.

The assumed crosswind resolution Δy is selected as follows: at low spatial resolution ($\Delta x \geq 10$ m), $\Delta y = \Delta x$; at high spatial resolution ($\Delta x < 10$ m), Δy is either equal to 10 m (single unit treatment) or equal to Δx (multiple discrete cells treatment).

Thus, looking at the 7-m-resolution case in Fig. 4, and assuming a single unit treatment, the first structure located at $x = 0$ has an effective value $SS = 7$ m (instead of the “true” value of 10 m), which corresponds to a compressed structure characterized by a footprint equal to $(7 \times 10) = 70$ m² (instead of a true value of 100 m²) and a peak heat release rate equal to $400 \text{ kW/m}^2 \times 70 \text{ m}^2 = 28$ MW (instead of a true value of 40 MW). The generation rates and travel distances of firebrands are reduced accordingly. Similarly, the effective separation distance between the first two structures is $SSD = 14$ m (instead of 10 m), which corresponds to an enlarged size and a reduced ability to collect firebrands from upwind burning structures. In this case, these modifications combine to artificially reduce the simulated risk of fire spread.

Assuming now a multiple discrete cells treatment, the first structure is characterized by a footprint equal to $(7 \times 7) = 49 \text{ m}^2$ (instead of a true value of 100 m^2) and a peak heat release rate equal to $400 \text{ kW/m}^2 \times 49 \text{ m}^2 = 19.6 \text{ MW}$ (instead of a true value of 40 MW). And the simulated risk of fire spread is even further reduced.

Furthermore, looking at the 30-m-resolution case in Fig. 4, one finds that the geometrical length scales SS and SSD are significantly modified. For instance, for $\Delta x = 30 \text{ m}$, the first structure located at $x = 0$ has an effective value $SS = 30 \text{ m}$ (instead of the “true” value of 10 m), which corresponds to an enlarged structure characterized by a footprint equal to $(30 \times 30) = 900 \text{ m}^2$ (instead of a true value ranging between 100 m^2 and 400 m^2 , depending on the number of 10-m-size structures contained in a given 900-m^2 -cell) and a peak heat release rate equal to $400 \text{ kW/m}^2 \times 900 \text{ m}^2 = 360 \text{ MW}$ (instead of a true value ranging between 40 MW and 160 MW). The generation rates and travel distances of firebrands are increased accordingly. Similarly, the effective separation distance between adjacent structures is $SSD = 30 \text{ m}$ (instead of 10 m), which corresponds to an enlarged size and a reduced ability to collect firebrands from upwind burning structures. In this case, these modifications tend to compensate each other but also lead to inaccurate results in fire spread as documented below (Section 3.4). Note that for $\Delta x = 30 \text{ m}$, there is no difference between the single unit and multiple discrete cells treatments.

Some other cases (not shown here) have interesting features: for $\Delta x = 20 \text{ m}$ and 40 m , SSD is decreased to 0 and the effects of the firebreaks are ignored (see Table 1).

Thus, variations in spatial resolution lead to several difficulties due to mapping errors: coarse simulations ($\Delta x \geq SS, SSD$) feature inaccurate values of SS and SSD, which will increase or decrease the propensity for structure-to-structure fire spread (see Section 3.4); in contrast, fine simulations ($\Delta x < SS, SSD$) feature more accurate values of SS and SSD but will lose accuracy when treating structures as multiple discrete cells (instead of a single entity).

The numerical configuration has also the following features: the terrain is flat; the wind is constant and uniform, and blows into the positive x -direction at a velocity (evaluated at 6.1-m elevation) $u_{wind} = 17.9 \text{ m/s}$ (40 mph). The computational domain extends from $x = 0$ to $x = 1000 \text{ m}$. Combustion and firebrand emission are initiated at time $t = 0$, at the first structure located at $x = 0$, and fire spreads to structures located downwind due to spot ignition. To isolate the effects of firebrand-driven fire spread (the focus of the present study), spread mechanisms due to thermal radiation and flame/plume contact are simply ignored.

3.2. Representative results

We first consider the results obtained with $\Delta x = 10 \text{ m}$. As explained in the previous section, this level of spatial resolution allows an exact representation of the problem geometry, and the accuracy of simulation results are in that case primarily controlled by the accuracy of the physical models and are not affected by mapping errors.

Temporal resolution is not an issue in the present simulations since the time required for the fire to spread between adjacent structures is

Table 1

Numerical parameters in the series of one-dimensional MATLAB simulations: downwind grid spacing, Δx ; crosswind grid spacing, Δy ; effective structure separation distance, SSD; effective structure size, SS.

Parameters	Values
Δx (m)	1; 3; 5; 7; 10; 13; 15; 17; 20; 23; 25; 27; 30; 40; 50
Δy (m)	10; 10; 10; 10; 13; 15; 17; 20; 23; 25; 27; 30; 40; 50
SSD (m)	10; 9 or 12; 10; 7 or 14; 10; 13 or 26; 30; 34 or 51; 0; 46 or 69; 50; 27 or 54; 30; 0; 50
SS (m)	10; 9 or 12; 10; 7 or 14; 10; 13 or 26; 30; 17 or 34; 20; 46 or 69; 50; 27 or 54; 30; 40; 50

several hundreds of seconds long while the numerical time step Δt is on the order of 1 s . We use $\Delta t = CFL \times (\Delta x / U_{wind})$ with $CFL = 1$ [24].

Fig. 5 presents the main features of the 10-m-resolution case. Fig. 5(a) presents the spatial variations of the total number of accumulated firebrands on the ground, noted N_{∞} . The variations of N_{∞} show a repeated peak and trough pattern that can be explained by the succession of firebrand-emitting structures and firebreaks: for instance, the region located at $10 \leq x \leq 20 \text{ m}$ is a firebreak and receives firebrands emitted by the first burning structure (located at $0 \leq x \leq 10 \text{ m}$); the region located at $20 \leq x \leq 30 \text{ m}$ is the location of the second structure and receives a smaller amount of firebrands from the first burning structure; the region located at $30 \leq x \leq 40 \text{ m}$ is a firebreak and receives a larger amount of firebrands emitted by the first and second burning structures, etc. It is seen that at $x > 50 \text{ m}$, the variations of N_{∞} achieve a quasi-steady pattern corresponding to the simultaneous burning of 2 structures (as shown in Fig. 2, the maximum value of firebrand flight distance for a 40-MW structure fire is $X_{max} = 84 \text{ m}$, which allows for 4 structures to burn simultaneously; the firebrand accumulation requirement used in the spot fire ignition model reduces that value to 2).

The total number of firebrands emitted by the first structure can be readily calculated using the design fire curve presented in Fig. 1. We find $N_{emitted} = (10 \text{ pcs/s/MW}) \times (400 \text{ kW/m}^2 \times 100 \text{ m}^2) \times ((300/2) + 3600 + (300/2) \text{ s}) = 1.56 \times 10^6 \text{ pcs}$. The PDF of flight distance presented in Fig. 2 suggests that 33% of the firebrands fall on the second structure located downwind: $N_{\infty} = 0.33 \times N_{emitted} \approx 5.15 \times 10^5 \text{ pcs}$. These calculations are in excellent agreement with the value of N_{∞} found at $20 \leq x \leq 30 \text{ m}$ in Fig. 5(a).

Fig. 5(b) presents the spatial variations of the total time of ignition of the structures, $\tau = (TOA + t_{ign,small} + t_{ign,large})$ (where the subscript k has been dropped for clarity). This time is set to 0 inside the firebreaks. The figure confirms that the fire successfully spreads across the array of structures, albeit at a slow pace. The time τ can be calculated as follows. Let us consider again the first burning structure located at $0 \leq x \leq 10 \text{ m}$. This structure emits $N_{emitted}$ firebrands and a fraction $N_{\infty} \approx 5.15 \times 10^5 \text{ pcs}$ of those fall on the second structure located at $20 \leq x \leq 30 \text{ m}$. In the spot fire ignition model discussed in Section 2, small flame ignition at the second structure requires the accumulated mass of firebrands per unit ground surface, ψ , to be larger than a critical value $\psi^*(v_{air})$ given by Eq. (1). With $u_{wind} = 17.9 \text{ m/s}$, we calculate v_{air} at 2-m elevation using a velocity boundary layer profile proposed by Albin [34] and get $v_{air} \approx 1.1 \text{ m/s}$; we finally obtain $\psi^*(v_{air}) \approx 0.059 \text{ g/cm}^2$. Assuming that the average weight of a firebrand is 0.2 g , we find that the critical value for the number of accumulated firebrands is $N_{critical} \approx (0.059/0.2) \times A_{cell} = (0.295 \text{ pcs/cm}^2) \times (10^6 \text{ cm}^2) = 2.95 \times 10^5 \text{ pcs}$, where A_{cell} is the surface area of the target cell (in units of cm^2). The fact that $N_{\infty} > N_{critical}$ confirms successful fire spread. Next, the time for ignition of a small flame, $t_{ign,small}$, may be calculated using the design fire curve (Fig. 1) through the following equation: $N_{critical} = 0.33 \times (10 \text{ pcs/s/MW}) \times (400 \text{ kW/m}^2 \times 100 \text{ m}^2) \times (t_{critical} - (300/2) \text{ s})$, where the $(300/2)$ term represents the effect of the initial linear growth phase; we get $t_{critical} \approx 2362 \text{ s}$. As explained in Ref. [24], ignition of a small flame also requires and additional 42-s delay and we finally have $t_{ign,small} \approx 2362 + 42 = 2404 \text{ s}$. The total time of ignition of the second structure can now be calculated as $\tau = (TOA + t_{ign,small} + t_{ign,large}) = (((SS + SSD)/U_{wind}) + t_{ign,small} + t_{ign,large}) = ((20/17.9) + 2404 + 300) = 2705 \text{ s}$. These calculations are in excellent agreement with the value of τ found at $20 \leq x \leq 30 \text{ m}$ in Fig. 5(b). Note that ignition of downwind structures is more rapid because these structures receive firebrands emitted from two upwind structures instead of one.

Thus, the rate of spread is relatively slow. For the initial event corresponding to the fire spreading from the first to the second structure, we have $ROS \approx ((SS + SSD)/\tau) = (20/2705) \approx 0.0074 \text{ m/s}$. As mentioned above, the mean value of the rate of spread, $\bar{ROS}$, is higher than this estimate because other structure-to-structure fire spread events feature multiple structures burning simultaneously (two structures in

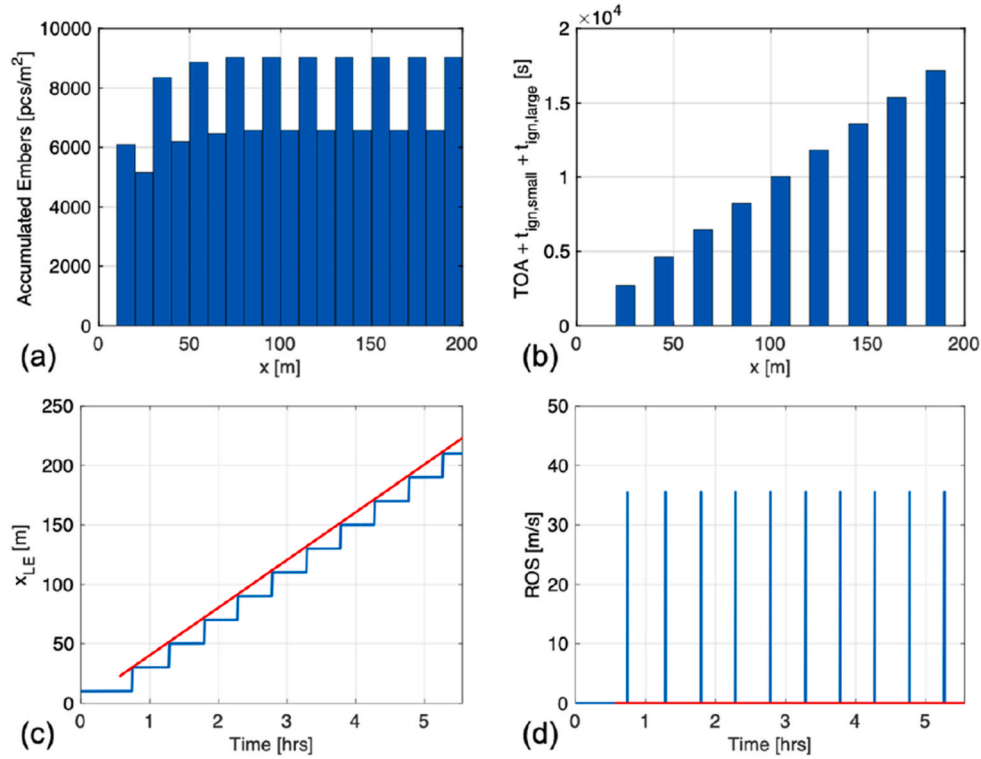


Fig. 5. Simulation of structure-to-structure fire spread with MATLAB. Spatial variations of: (a) the total number of accumulated firebrands on the ground; (b) the time of ignition of the fire, $(TOA + t_{ign,small} + t_{ign,large})$. Temporal variations of: (c) the fire leading edge position, x_{LE} ; (d) the fire rate of spread, ROS. The red solid lines in figures (c)-(d) correspond to the time-averaged solution, $\bar{x}_{LE}(t) = (\bar{ROS} \times t)$. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

the present case).

A different perspective is adopted in Fig. 5(c) which presents the temporal variations of the fire leading edge position, $x_{LE}(t)$. The plot shows the long periods of no spread separated by discrete events corresponding to the sudden ignition of a new structure. In the present case, on average, it takes approximately 5 h for the fire to spread across an area of 200 m.

Fig. 5(d) presents the corresponding temporal variations of the rate of spread, ROS. The rate of spread can be directly extracted from the results presented in Fig. 5(c) and calculated as the time derivative of the variations $x_{LE}(t)$, $ROS = d(x_{LE})/dt$. Fig. 5(d) shows that due to the alternating periods of spread and no spread, the instantaneous values of simulated ROS fluctuate between the value of 0 and a peak value equal to $((SS + SSD)/\Delta t) = ((SS + SSD)/(CFL \times (\Delta x/U_{wind}))) = (20/(1 \times (10/17.9))) \approx 35.8$ m/s. The time-averaged value is $\bar{ROS} \approx 0.01$ m/s.

Note that the fire spread in the academic test presented in Fig. 5 is quite regular. In real fire disasters, structure-to-structure fire spread features significant variability. ELMFIRE will capture some of the sources of variability through variations in the wind conditions, the building details, the separation distance between buildings, the possible presence of fuel between structures, etc.

3.3. Maps of fire spread conditions

We further explore in this section the predictions of the firebrand model and consider an analysis of the critical conditions that lead to structure-to-structure fire spread, in terms of separation distance, SSD, wind velocity, u_{wind} , and firebrand generation rate, GR. We chose to represent the effect of the firebrand generation rate through the ratio (GR/HRR) (in units of pcs/s/MW). All results are here obtained with $SS = 10$ m and $\Delta x = 1$ m (1-m-resolution is used for an exact representation of all values of SSD).

Fig. 6 presents different representations of the conditions leading to successful or failed structure-to-structure fire spread. Each marker in Fig. 6(a) and 6(b) represents one MATLAB simulation. In this exploration of the parameter space, SSD takes values between 2.5 and 30 m, and u_{wind} takes values between 4.5 and 67 m/s (between 10 and 150 mph). As may have been expected, Fig. 6(a) shows that fire spread between structures is promoted by lower values of the separation distance and larger values of the firebrand generation rate. Fig. 6(b) also shows that there is a finite range of wind velocities that lead to successful fire spread: critically low values of u_{wind} do not provide sufficient mass transfer of oxygen to sustain smoldering combustion; critically high values of u_{wind} provide excessive convective cooling.

While the results presented in Fig. 6 clearly call for a careful validation study prior to drawing quantitative conclusions, they already provide valuable qualitative information and serve to illustrate the potential of the firebrand model in the calculation of fire behavior in WUI/urban communities.

3.4. Sensitivity to changes in spatial resolution

We consider again in this section the baseline configuration ($SS = SSD = 10$ m) and analyze the sensitivity of the firebrand model to changes in spatial resolution. Section 3.1 already discussed the effects of changing the computational grid spacing Δx : coarse simulations ($\Delta x \geq SS, SSD$) feature inaccurate values of SS and SSD, while fine simulations ($\Delta x < SS, SSD$) lose accuracy when treating structures as multiple discrete cells.

Fig. 7 provides a quantification of these effects by plotting the mean rate of fire spread, $\bar{ROS}$, as a function of changes in the computational grid spacing, Δx . $\bar{ROS}$ is calculated as the time-averaged value of the variations $x_{LE}(t)$ during the steady state phase of the solution. Fig. 7 compares simulation results obtained with the single unit treatment and

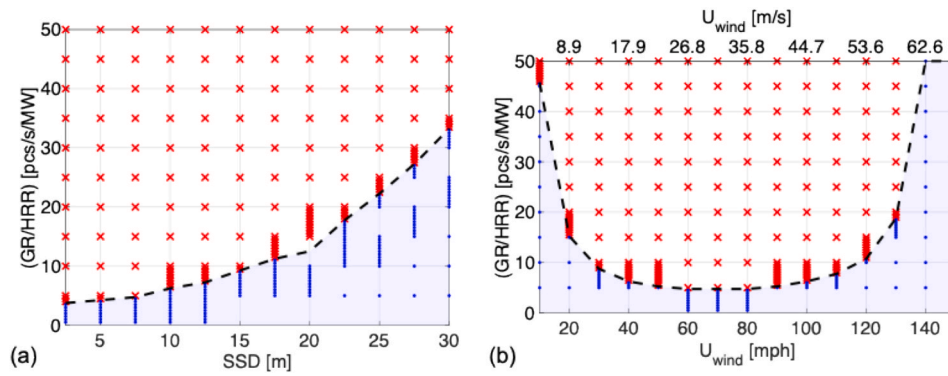


Fig. 6. Simulation of structure-to-structure fire spread with MATLAB. Maps showing conditions of structure-to-structure fire spread as a function of: (a) SSD and (GR/HRR) with $u_{wind} = 17.9$ m/s; (b) u_{wind} and (GR/HRR) with $SSD = 10$ m. Red crosses mark conditions of fire spread; blue dots mark conditions of no fire spread. As a reminder, fire spread here is due to the sole effects of firebrands. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

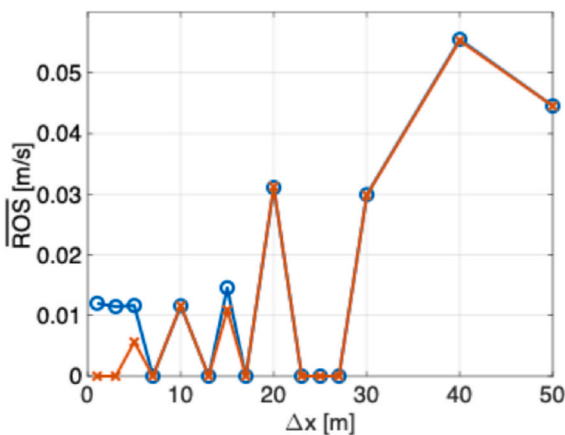


Fig. 7. Simulation of structure-to-structure fire spread with MATLAB. Variations of the mean rate of fire spread, $\overline{ROS}$, with the computational grid spacing, Δx . The blue (red) solid line with circle (cross) symbols correspond to results obtained using the single unit treatment (multiple discrete cells treatment). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

the multiple discrete cells treatment.

Fig. 7 shows that the firebrand model is quite sensitive to changes in spatial resolution. Let us first consider results obtained using the single unit treatment. While simulations performed with $\Delta x = 1, 3$ and 5 m provide similar results ($\overline{ROS} \approx 0.01$ m/s), simulations performed at coarser resolutions provide scattered results corresponding to no spread ($\overline{ROS} = 0$) when $\Delta x = 7, 13, 17, 23, 25$ and 27 m, or to accelerated spread ($\overline{ROS} > 0.01$ m/s) when $\Delta x = 20, 30, 40$ and 50 m. Compared to its value obtained with $\Delta x = 10$ m, $\overline{ROS}$ is three times larger when $\Delta x = 30$ m (the baseline resolution of ELMFIRE) and more than five times larger when $\Delta x = 40$ m. As discussed in Section 3.1, the simulation performed with $\Delta x = 10$ m features an exact representation of the problem geometry; this simulation gives $\overline{ROS} \approx 0.01$ m/s and deviations from that value observed in Fig. 7 are due to mapping errors that first lead to inaccurate effective values of SS and SSD and then to an inaccurate value of $\overline{ROS}$.

Thus, when using the single unit treatment, the results are grid-converged (i.e., approximately independent of changes in Δx) provided that $\Delta x \leq 5$ m, or equivalently $\Delta x \leq (SS/2), (SSD/2)$. These results can be explained by the dominant role of SS and SSD in the WUI/urban fire problem: both length scales need to be captured by the computational grid for the solution to be “numerically accurate” (i.e.,

independent of the choice of numerical parameters). As explained in Refs. [23,24], a “numerically accurate” solution refers to a solution that is controlled by physical rather than numerical parameters; the solution may still be “physically inaccurate” but physical parameters are the dominant factor in the model performance.

Let us now consider results obtained using the multiple discrete cells treatment. When $\Delta x \geq SS, SSD$, these results are similar to those obtained using the single unit treatment; when $\Delta x < SS, SSD$, they differ significantly and are less accurate than those obtained using the single unit treatment. For instance, Fig. 7 shows that simulations performed with $\Delta x = 1$ and 3 m feature no fire spread.

In conclusion, the results of the firebrand model are strongly affected by the choice of spatial resolution and are grid-dependent when $\Delta x \geq SS, SSD$ and/or when using the multiple discrete cells treatment (the current treatment in ELMFIRE). Numerical accuracy requires both high spatial resolution, $\Delta x \leq (SS/2), (SSD/2)$, and the single unit treatment.

4. Two-dimensional structure-to-structure fire spread in ELMFIRE

We now turn to a final series of tests performed using ELMFIRE [21]. The configuration is identical to that considered in Section 3 and corresponds to one-dimensional fire spread driven by the transport of firebrands in the wind direction but is here treated in a more realistic two-dimensional framework. The numerical layout of structures is pre-processed in GDAL [38] using mode resampling at the desired target resolution before being imported into ELMFIRE. These tests allow a verification that the MATLAB- and ELMFIRE-based firebrand models provide identical results.

The computational domain extends from $x = 0$ to $x = 1200$ m in the wind x -direction, and from $y = 0$ to $y = 400$ m in the cross-flow y -direction; the initial fire corresponds to the line of structures located at $x = 1200$ m. Fig. 8 presents a small section of the computational domain and illustrates the changes in the geometrical footprints of structures that result from mapping errors. These errors are similar to those presented in Fig. 4.

Consistent with data presented in Section 3.1, Fig. 8 shows that at high spatial resolution ($\Delta x = \Delta y = 7$ m), the geometrical length scales SS and SSD are approximately captured by the computational grid and their effective values are close to the true values: the geometrical footprint of structures ranges from $(7 \times 7) = 49 \text{ m}^2$ to $(14 \times 14) = 196 \text{ m}^2$ (instead of the true value of 100 m^2) and because of the multiple discrete cells treatment used in ELMFIRE, the effective footprint used by the firebrand algorithm is $(\Delta x \times \Delta y) = 49 \text{ m}^2$. The effective separation distance is 7 or 14 m.

Fig. 8 also shows that at low spatial resolution ($\Delta x = \Delta y = 30$ m),

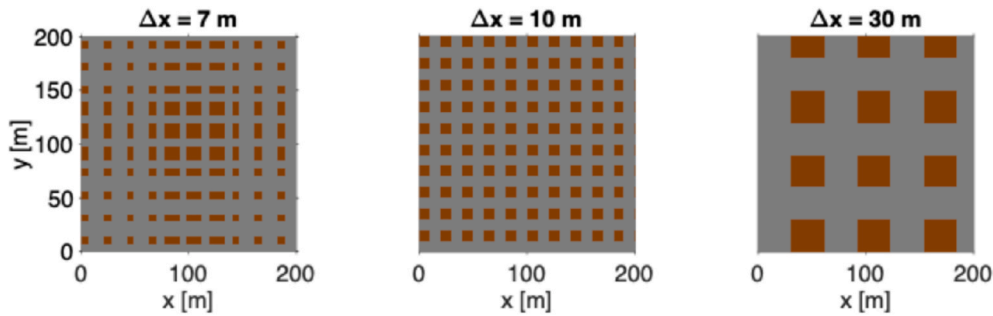


Fig. 8. Numerical configuration used in the two-dimensional ELMFIRE tests and corresponding to an array of alternating structures (colored in brown) and firebreaks (colored in grey). The left (right) figure corresponds to fine-resolution (coarse-resolution), $\Delta x = \Delta y = 7$ m ($\Delta x = \Delta y = 30$ m); the central figure corresponds to $\Delta x = \Delta y = 10$ m and shows the exact geometry of the problem in which $SS = SSD = 10$ m. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

the geometrical length scales SS and SSD are strongly modified by mapping errors: the effective footprint of structures is $(30 \times 30) = 900$ m² (instead of a true value ranging between 100 m² and 400 m², depending on the number of 10-m-size structures contained in a given 900-m²-cell). The effective separation distance is 30 m.

Fig. 9 provides a quantification of these effects by plotting the mean rate of fire spread, $\overline{ROS}$, as a function of changes in the computational grid spacing, $\Delta x = \Delta y$. The results presented in Fig. 9 are close to those presented in Fig. 7 (using the multiple discrete cells treatment); slight discrepancies are due to small differences in the post-processing treatment of the MATLAB and ELMFIRE results. Fig. 9 confirms the absence of grid convergence in the fire spread model: coarse simulations ($\Delta x \geq SS, SSD$) feature inaccurate values of SS and SSD , while fine simulations ($\Delta x < SS, SSD$) lose accuracy when treating structures as multiple discrete cells.

5. Summary of current model limitations

As explained at the end of the introduction (section 1), the objective of the present study is to evaluate the performance of a new firebrand model proposed for landscape-scale fire spread simulators. The model performance is evaluated in simple academic-like verification tests with a focus on grid-convergence, i.e., on the sensitivity of the results to changes in spatial resolution.

Now, in addition to reducing numerical errors, a model development strategy requires reducing physical modeling errors. We recognize that the present firebrand model cannot be yet considered as a tested and validated predictive capability. The following is a list of current

limitations and open questions that will require additional analysis and/or validation studies: (1) the firebrand generation model assumes a linear relationship between generation rate and heat release rate that should be tested and enhanced to account for additional effects; (2) the firebrand transport model uses PDFs for flight distance that have been developed under a limited set of conditions and that should be tested under a wider range of conditions; (3) the firebrand transport model does not explicitly account for the firebrand shape and size distributions; (4) the firebrand transport model currently relies on a simple treatment for flight time that assumes a horizontal trajectory and does not include the effects of lofting, gravity, and vertical variations in wind velocity; (5) the wind velocity boundary layer profile assumed in ELMFIRE is static and does not account for expected modifications of the local wind conditions by the structures and by the burning fuels; (6) the firebrand ignition model does not include an explicit treatment of surface creep and accumulation in the vicinity of structures; (7) the firebrand ignition model does not consider the effects of different target fuels, variable moisture content, and different fire penetration scenarios; (8) the firebrand ignition model uses an empirical t -square-law description of the time delay associated with the transition from a small flame to a fully-developed fire that should be tested; (9) the firebrand ignition model does not consider the effect of cooling and burnout of firebrands that accumulate around structures.

Future work will be aimed at addressing these limitations.

6. Conclusions

This study presents a detailed evaluation of the performance of a recently developed firebrand model in a new application to WUI/urban fires. The study is part of a modeling effort aimed at developing an accurate treatment of firebrand effects in an established landscape-scale fire spread simulator called ELMFIRE. The firebrand model includes physical descriptions of generation, transport and spot ignition, and was previously evaluated in applications to wildland fires [23,24]. The present study focuses on firebrand-driven urban fire spread, considers a benchmark configuration corresponding to an array of identical, equally-spaced, square structures, and includes simulations performed in one dimension using MATLAB and in two dimensions using ELMFIRE.

The benchmark configuration features two characteristic length scales: the structure size (SS) and structure separation distance (SSD); we use $SS = SSD = 10$ m. When the geometry of the problem is correctly captured by the computational grid, i.e. when the effective values of SS and SSD are accurate (for instance, when $\Delta x = SS = SSD = 10$ m) and when structures are treated by the model as single entities, it is shown that the firebrand model provides physically-meaningful results that correctly reproduce (at least qualitatively) expected effects of separation distances and wind velocities. While beyond the scope of the present study, we recognize that to become a trusted quantitative tool for fire risk calculations, the firebrand model still requires validation

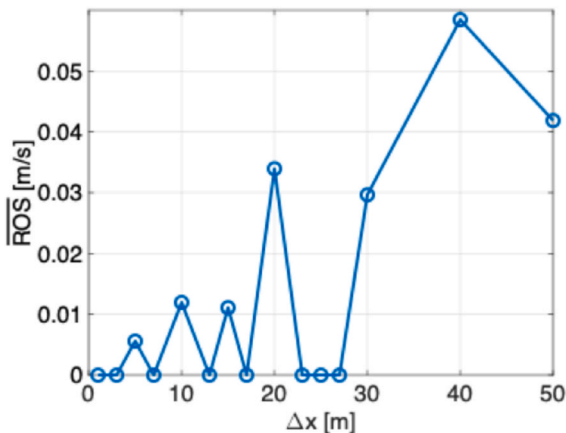


Fig. 9. Simulation of structure-to-structure fire spread with ELMFIRE. Variations of the mean rate of spread of the fire, $\overline{ROS}$, with the computational grid spacing, Δx .

studies in which simulation results are systematically compared to data obtained from field-scale fire tests and past fire disasters (see section 5).

The present study also reveals a separate issue found when the geometry of the problem is not correctly captured by the computational grid. Landscape-scale fire models typically fix the spatial resolution to a baseline level (30 m in ELMFIRE) and are calibrated to work at that particular baseline resolution. To the best of our knowledge, landscape-scale fire models do not consider the sensitivity of their results to changes in spatial resolution. While this may not be a concern in wildland fire applications where the length scales of the problem (i.e., the length scales associated with changes in the topography, vegetation coverage, and wind conditions) are relatively large (several 10s of meter), this becomes a concern in WUI/urban fire applications where the length scales of the problem are much smaller (SS and SSD are typically around 5 to 10 m). The present results suggest that the mapping errors associated with an inaccurate representation of the community layout lead to very large errors in the predicted fire behavior, ranging from fires that fail to spread when the correct behavior is a spreading fire, to fires that spread more than five times faster than the correct rate-of-spread value. These mapping errors are of two kinds: inaccurate effective values of SS and SSD when using coarse resolution, $\Delta x \geq SS$, SSD; or an inaccurate treatment of structures (as multiple discrete entities instead of single entities) when using fine resolution, $\Delta x < SS$, SSD. The present results suggest that numerical accuracy requires both high spatial resolution and a treatment of structures as the relevant unit in the fire spread problem (referred to above as the single unit treatment). In conclusion, “urban conflagration solvers” should work at a higher resolution (5 or 10 m) than that currently used.

CRedit authorship contribution statement

Yiren Qin: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Dwi M.J. Purnomo:** Software, Methodology, Investigation. **Maria Theodori:** Software, Methodology, Investigation. **Maryam Zamanialaei:** Software, Methodology, Investigation. **Chris Lautenberger:** Supervision, Software, Resources, Project administration, Investigation, Funding acquisition, Conceptualization. **Michael Gollner:** Supervision, Resources, Project administration, Conceptualization. **Arnaud Trouvé:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This project is financially supported by the Gordon and Betty Moore Foundation. Previous financial support by the US National Science Foundation (NSF project entitled “PREEVENTS Track 2: Fire Spread at the Wildland-Urban Interface Modeling and Data Assimilation for Prediction and Risk assessment”; Award number 1854952) is gratefully acknowledged.

References

- [1] Fire Dynamics Simulator (FDS), developed by the National Institute of Standards and Technology, <https://pages.nist.gov/fds-smv/> (accessed September 9, 2025).
- [2] OpenFOAM, developed by OpenCFD Ltd. <https://www.openfoam.com>. (Accessed 9 September 2025).

- [3] FireFOAM, developed by FM. <https://github.com/fireFoam-dev>. (Accessed 9 September 2025).
- [4] W. Mell, M.A. Jenkins, J. Gould, P. Cheney, A physics-based approach to modeling grassland fires, *Int. J. Wildland Fire* 16 (2007) 1–22, <https://doi.org/10.1071/WF06002>.
- [5] N. Masoudvaziri, F. Szasdi Bardales, O. Kaan Keskin, A. Sarreshtehdari, K. Sun, N. Elhami-Khorasani, Streamlined wildland-urban interface fire tracing (SWUIFT): modeling wildfire spread in communities, *Environ. Model. Software* 143 (2021) 105097, <https://doi.org/10.1016/j.envsoft.2021.105097>.
- [6] F. Szasdi-Bardales, K. Shamsaei, N.P. Lareau, T.W. Juliano, B. Kosovic, H. Ebrahimian, N. Elhami-Khorasani, Integrating dynamic wildland fire position input with a community fire spread simulation: a case study of the 2018 camp fire, *Fire Saf. J.* 143 (2024) 104076, <https://doi.org/10.1016/j.firesaf.2023.104076>.
- [7] H. Mahmoud, A. Chulahwat, Unraveling the complexity of wildland urban interface fires, *Sci. Rep.* 8 (2018) 9315, <https://doi.org/10.1038/s41598-018-27215-5>.
- [8] A. Chulahwat, H. Mahmoud, S. Monedero, F.J. Diez Vizcaino, J. Ramirez, D. Buckley, A. Cardil Forradellas, Integrated graph measures reveal survival likelihood for buildings in wildfire events, *Sci. Rep.* 12 (2022) 15954, <https://doi.org/10.1038/s41598-022-19875-1>.
- [9] A. Chulahwat, H. Mahmoud, The impact of wind characteristics on the spatial distribution of damage to the built environment during wildfire events: the 2022 Marshall fire, *Nat. Hazards Rev.* 25 (2024) 06023003, <https://doi.org/10.1061/NHREFO.NHENG-1888>.
- [10] D.M.J. Purnomo, Y. Qin, M. Theodori, M. Zamanialaei, C. Lautenberger, A. Trouvé, M.J. Gollner, Integrating an urban fire model into an operational wildland fire model to simulate one dimensional wildland-urban interface fires: a parametric study, *Int. J. Wildland Fire* 33 (2024), <https://doi.org/10.1071/WF24102>.
- [11] D.M.J. Purnomo, Y. Qin, M. Theodori, M. Zamanialaei, C. Lautenberger, A. Trouvé, M.J. Gollner, Reconstructing modes of destruction in wildland-urban interface fires using a semi-physical level-set model, *Proc. Combust. Inst.* 40 (2024) 105755, <https://doi.org/10.1016/j.proci.2024.105755>.
- [12] M. Hamada, On the rate of fire spread, *Dis. Res.* 1 (1951) 35–44. Non-life insurance rating organization of Japan, (in Japanese).
- [13] W. Jiang, F. Wang, L. Fang, X. Zheng, X. Qiao, Z. Li, Q. Meng, Modelling of wildland-urban interface fire spread with the heterogeneous cellular automata model, *Environ. Model. Software* 135 (2021) 104895, <https://doi.org/10.1016/j.envsoft.2020.104895>.
- [14] E. Mastorakos, S. Gkantonas, G. Efstathiou, A. Giusti, A hybrid stochastic Lagrangian –cellular automata framework for modelling fire propagation in inhomogeneous terrains, *Proc. Combust. Inst.* 39 (2023) 3853–3862, <https://doi.org/10.1016/j.proci.2022.07.240>.
- [15] Y. Cao, J. Jiang, H. Sun, W. Chen, J. Ye, Modeling of urban fire spread with integrated cellular automata (CA)-unity framework: case study on 2025 wildfire in Altadena, *Sustain. Cities Soc.* 134 (2025) 106961, <https://doi.org/10.1016/j.scs.2025.106961>.
- [16] K. Himoto, T. Tanaka, Development and validation of a physics-based urban fire spread model, *Fire Saf. J.* 43 (2008) 477–494, <https://doi.org/10.1016/j.firesaf.2007.12.008>.
- [17] S.W. Lee, R.A. Davidson, Physics-based simulation model of post-earthquake fire spread, *J. Earthq. Eng.* 14 (2010) 670–687, <https://doi.org/10.1080/13632460903336928>.
- [18] N. de Koker, R.S. Walls, A. Cicione, Z.R. Sander, S. Löffel, J.J. Claasen, S.J. Fourie, L. Croukamp, D. Rush, 20 dwelling large-scale experiment of fire spread in informal settlements, *Fire Technol.* 56 (2020) 1599–1620, <https://doi.org/10.1007/s10694-019-00945-2>.
- [19] A. Cicione, C. Wade, M. Spearpoint, L. Gibson, R. Walls, D. Rush, A preliminary investigation to develop a semi-probabilistic model of informal settlement fire spread using B-RISK, *Fire Saf. J.* 120 (2021) 103115, <https://doi.org/10.1016/j.firesaf.2020.103115>.
- [20] S. Stevens, D. Rush, Urban fire spread modelling: a review of dynamic computational models and potential for application to informal settlement fires, *Int. J. Disaster Risk Reduct.* 124 (2025) 105528, <https://doi.org/10.1016/j.ijdrr.2025.105528>.
- [21] A. Elmfire, Eulerian level set model of FIRE spread. <https://elmfire.io>. (Accessed 9 September 2025).
- [22] C. Lautenberger, Wildland fire modeling with an Eulerian level set method and automated calibration, *Fire Saf. J.* 62 (2013) 289–298, <https://doi.org/10.1016/j.firesaf.2013.08.014>.
- [23] Y. Qin, A. Trouvé, A numerically accurate mathematical framework for simulations of firebrand transport in landscape-scale fire spread models, *Fire Saf. J.* 152 (2025) 104343, <https://doi.org/10.1016/j.firesaf.2025.104343>.
- [24] Y. Qin, A. Trouvé, Simulations of firebrand ignition in landscape-scale fire spread models, *Fire Saf. J.* (2025) submitted for publication.
- [25] E. Koo, P.J. Pagni, D.R. Weise, J.P. Woycheese, Firebrands and spotting ignition in large-scale fires, *Int. J. Wildland Fire* 19 (2010) 818–843, <https://doi.org/10.1071/WF07119>.
- [26] S.E. Caton, R.S.P. Hakes, D.J. Gorham, A. Zhou, M.J. Gollner, Review of pathways for building fire spread in the wildland urban interface part I: exposure conditions, *Fire Technol.* 53 (2017) 429–473, <https://doi.org/10.1007/s10694-016-0589-z>.
- [27] A.I. Filkov, V. Tihay-Fellicelli, N. Masoudvaziri, D. Rush, A. Valencia, Y. Wang, D. L. Blunck, M.M. Valero, K. Kempna, J. Smolka, J. De Beer, Z. Campbell-Lochrie, F. R. Centeno, M.A. Ibrahim, C.K. Lemmert, W.C. Tam, A review of thermal exposure and fire spread mechanisms in large outdoor fires and the built environment, *Fire Saf. J.* 140 (2023) 103871, <https://doi.org/10.1016/j.firesaf.2023.103871>.

- [28] S.L. Manzello, S. Suzuki, M.J. Gollner, A.C. Fernandez-Pello, Role of firebrand combustion in large outdoor fire spread, *Prog. Energy Combust. Sci.* 76 (2020) 100801, <https://doi.org/10.1016/j.pecs.2019.100801>.
- [29] N. Sardoy, J.L. Consalvi, A. Kaiss, A.C. Fernandez-Pello, B. Porterie, Numerical study of ground-level distribution of firebrands generated by line fires, *Combust. Flame* 154 (2008) 478–488, <https://doi.org/10.1016/j.combustflame.2008.05.006>.
- [30] K. Himoto, T. Tanaka, Transport of disk-shaped firebrands in a turbulent boundary layer, in: *Proc. 8th Intl. Symp. Fire Safety Science*, 8, 2005, pp. 433–444, <https://doi.org/10.3801/IAFSS.FSS.8-433>.
- [31] U. Dunder, S. Selamet, Fire load and fire growth characteristics in modern high-rise buildings, *Fire Saf. J.* 135 (2023) 103710, <https://doi.org/10.1016/j.firesaf.2022.103710>.
- [32] J.A. De Beer, Experimental Analysis and Numerical Modeling of Ignition of Lignocellulosic Building Material Subjected to Glowing Firebrand Piles, University of Maryland, College Park, USA, 2023, <https://doi.org/10.13016/dspace/qjr6-kxw3>. PhD thesis.
- [33] J.A. De Beer, E.L. Dietz, S.I. Stoliarov, M.J. Gollner, An empirical firebrand pile heat flux model, *Fire Saf. J.* 141 (2023) 104004, <https://doi.org/10.1016/j.firesaf.2023.104004>.
- [34] F.A. Albini, “Spot fire distance from burning trees-a predictive model,” USDA forest service intermountain forest and range experiment station general tech. Rep. Int. Ogden (1979). <https://www.frames.gov/catalog/8153>.
- [35] Y. Qin, A Physics-based Eulerian Framework for Modeling Firebrand Showering in regional-scale Wildland and wildland-urban Interface Fire Simulations, University of Maryland, College Park, USA, 2025, <https://doi.org/10.13016/skim-2dop>. PhD thesis.
- [36] X. Ju, M. Conkling, M. Hajilou, S. Lin, F. Mostafa, A. Ayyar, A. McDowell, M. Lisano, M.J. Gollner, Laboratory quantification of firebrand generation from WUI fuels for model development, *Fire Saf. J.* 141 (2023) 103921, <https://doi.org/10.1016/j.firesaf.2023.103921>.
- [37] K. Himoto, Chapter 8: firebrands, in: *Large Outdoor Fire Dynamics*, first ed., CRC Press, 2023.
- [38] GDAL/OGR geospatial data abstraction software library. <https://gdal.org>. (Accessed 9 September 2025).